2025/ 1 / 10

X

Preliminary Examination - 2025

Science and Technology (Part -I) (72)

Time - 2 hrs. **pages - 05** **Marks : 40**

Note : -
- **i) All questions are compulsory.**
- **ii) Use of calculator is not allowed.**
- **iii) The numbers to the right of the questions indicate full marks.**
- **iv) In case of MCQs (Q.No.1(A) only the first attempt will be evaluated and will be given credit.**
- **v) For every MCQ, the correct alternative (A),(B),(C) or (D) with subquestion number is to be written as an answer. for Eg- i) (A), ii) (B) , iii) (C)**
- **vi) Scientifically correct,labelled diagrams should be drawn wherever necessary.**

Q.1 A) Choose the correct alternatives and write correct option. **05**

i) In practice the unit .....is used for the measurement of electrical energy.

A) Joule B) Arge

C) Volt D) KWh

ii) If the vefractive index of glass with respect to air is 3/2, what is the refractive index of air with respect to glass?

A) $\frac{1}{2}$ B) 3 C) $\frac{1}{3}$ D) $\frac{2}{3}$

iii) If the focal length of a convex lens is 20 cm then its power is .....

A) 0.20 D B) 0.5D

C) -5.0D D) + 5.0 D

iv) Chemically noble gases are .....

A) more effective B) neutral

C) medium effective D) less effective

v) Availability of television,telephone and internet services all over the nation has been possible due to the satellite.

A) PSLV B) GSLV

C) IRS D) GSAT

B) Answer the following questions as per the given instructions. **05**

i) Find the odd man

Electric Moter, galvano meter, Voltmeter, Electric generator.

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B) ii) Complete the co-relation

Fe^{2+} : Ferrous : : Fe^{3+}: ----------

iii) Match the correct pair :

Column 'A' position of the object	Column 'B' position of the image
a) At 2F,	a) At focus F_2 b) At $2F_2$ c) Beyond $2F_2$

iv) State True or False

The conical cells respond to the colour.

v) What is meant by an artificial satellite?

Q2. A) Give Scientific reasons.(any two) **04**

i) Tungsten metal is used to make a solenoid type coil in an electric bulb.

ii) Copper pots pabinates in humid air.

iii) Atomic radius goes on increasing down a group.

B) Answer the following questions. (Any three) **06**

i) If water is heated for 0°c.What will you observe about its volume in following conditions.

a) If the temperature is increased up to 4°c.What will happen about its volume?

b) If the temperature is increased beyond 4°c.What will happen about its volume?

ii) Draw two structural formulae having molecular formula C_4H_{10}

iii) Write the first and second laws given by Kepler.

iv) Complete the following table.

Types of satellite	The names of Indian satellite and launcher
a) Weather satellite	satellite : ..... Launcher: .....
b) Communication satellite	satellite : ..... Launcher: .....

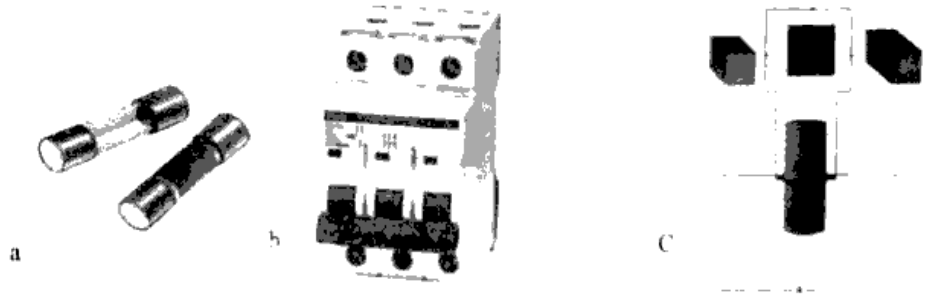
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v) Complete the following table.

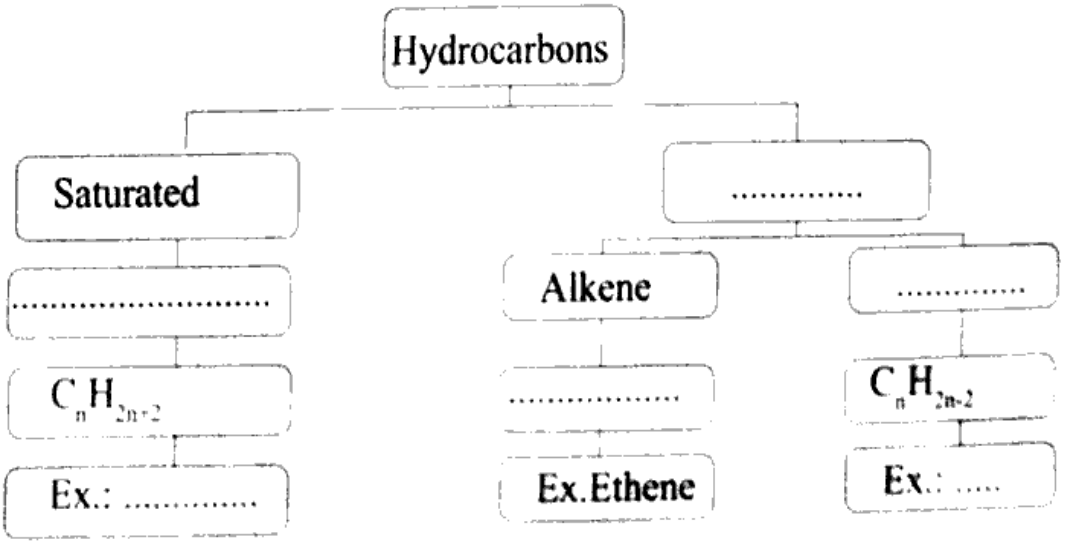
Processes /reaction	Endothermic / Exothermic
1) Melting of ice	.....
2) Dissolution of potassium nitrate in water.	.....
3) Decomposition of calcium carbonate.	.....
4) Dilution of conc.H_2SO_4 with H_2O	.....
5) Formation of ice from water.	.....
6) Dissolution of sodium hydroxide with water	.....

vi) What is meant by one Kcal heat? Write SI units and CGS units of heat.

vii) Identify the figures and explain their use.



viii) Complete the following flow chart :



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B) v) $_{13}AI$, $_{14}Si$, $_{11}Na$, $_{12}Mg$, $_{6}C$, $_{3}Li$, $_{9}F$, $_{7}N$

Which of the above elements has the highest metallic character and highest nonmetallic character?

Q.3 Answer the following questions.(any five) **15**

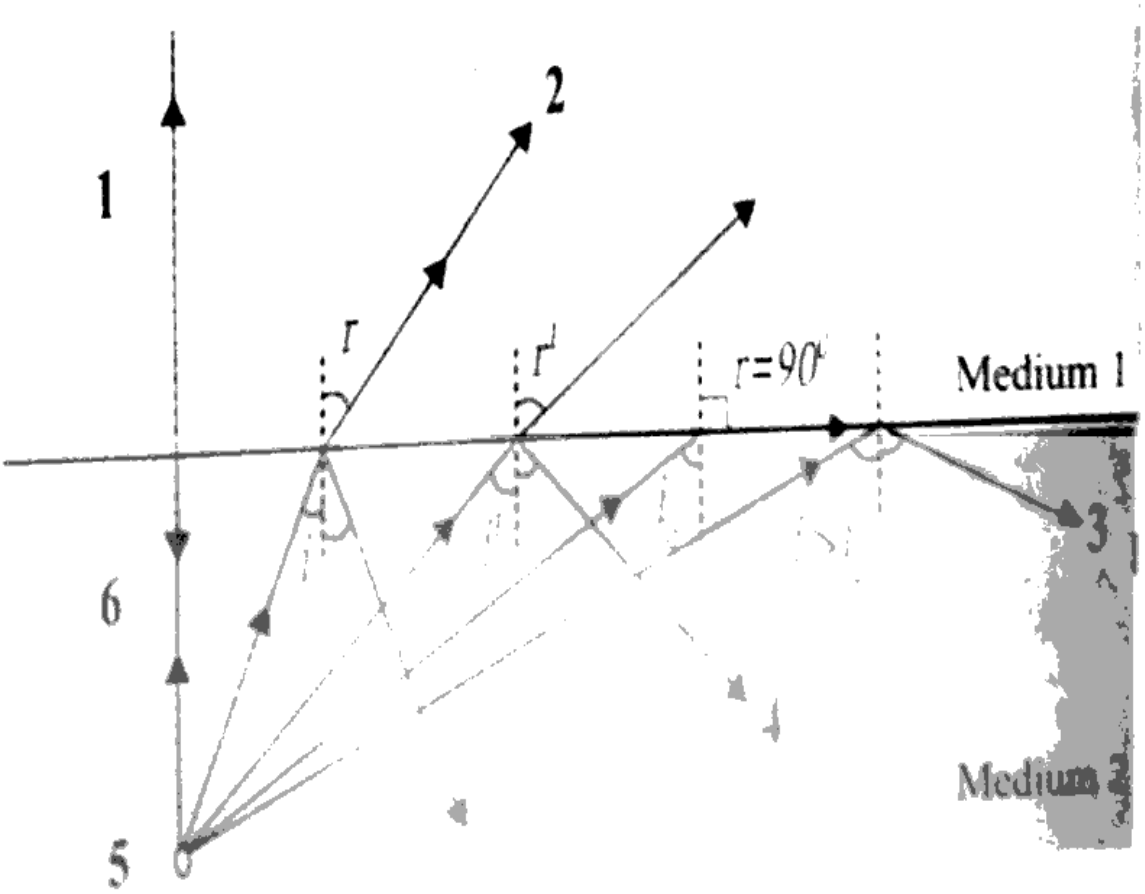
i) A tennis ball is thrown up and reaches a height of 4.05m before coming down.What was its initial velocity? How much total time will it take to come down? Assume g = 10m/s^2

ii) Atomic mass of 'X' is 6.9 and that of 'Y' is 39.1 then mean Z of X and Y is 23.

a) Which law is followed in above information?

b) Write the names of X,Y and Z.

iii) Observe the figure and label it at numbers 1 to 6.



iv) a) Write an example of reaction of nonmetals with oxygen to form neutral oxide.

[illegible] oxide or not? is there any exception

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Q.4 Answer any one of the following questions. **05**

1) Answer the following questions regarding concave lens.

a) In which equipments concave lens is used? 01

b) In the peep hole in door how many concave lens are used? 01

c) Concave lens are used in spectacles, Why? 01

d) Why concave lens is used in torch? 02

2) Complete the following table.

	parent name	structural formula	straight chain
1)	ethane	.....	C - C
2)	.....	CH_3-CH=CH_2	C - C = C
3)	butane	.....	C- C- C- C
4)	propane	CH_3-CH_2-COOH	.....
5)	Propene	.....	C - C ≡ C